

Alia 4 - Beta Alia eVTOL LED Animation Controller

LED animation controller for a scale model of the Beta Alia eVTOL aircraft. Features realistic flight simulation with dynamic prop and tail animations.

Key Documents

Document	Description	QR Code
GitHub Repo (https://github.com/johncohn/alia_blinky)	Top-level repository — browse, download, or fork	
TUTORIAL.md (https://github.com/johncohn/alia_blinky/blob/main/TUTORIAL.md)	Complete setup and customization guide	
SOLDER.md (https://github.com/johncohn/alia_blinky/blob/main/SOLDER.md)	Photo guide: soldering & gluing the PCB	

Version: 1.2.0 **Build:** 16 **Hardware:** Supports multiple boards (auto-detected) + 41 WS2812B LEDs

- Seeed XIAO RP2040
- Adafruit QT Py RP2040
- Seeed XIAO ESP32-S3
- Adafruit QT Py ESP32-S3 (no PSRAM)

Features

Auto-cycles through 4 animation patterns:

1. **FLIGHT** (~36s) - Realistic eVTOL flight simulation with 5 phases
 - Lift, transition in, conventional flight, transition out, landing
 - Non-linear prop acceleration/deceleration
 - Counter-rotating props, variable tail speed
2. **SLOW RAINBOW** (10s) - Full rainbow color cycle

3. **FAST WHITE** (10s) - Running white lights with sine wave
4. **RAINBOW PROPS** (10s) - Theater chase rainbow effect

Quick Start

Requirements

- Arduino IDE 2.x
- One of the supported boards (see Hardware section below)
- Adafruit NeoPixel library

Setup (Arduino IDE)

1. Install Board Support

For RP2040 boards:

- Tools → Board → Boards Manager
- Search "rp2040"
- Install "Raspberry Pi Pico/RP2040" (v5.4.3+)
- Select your board:
 - Tools → Board → Seeed XIAO RP2040, or
 - Tools → Board → Adafruit QT Py RP2040

For ESP32-S3 boards:

- Tools → Board → Boards Manager
- Search "esp32"
- Install "esp32 by Espressif Systems" (v3.3.3+)
- Select your board:
 - Tools → Board → esp32 → XIAO_ESP32S3, or
 - Tools → Board → esp32 → Adafruit QT Py ESP32-S3 no psram

2. Install Library

- Tools → Manage Libraries
- Search "Adafruit NeoPixel"
- Install (v1.15.2+)

3. Upload

- Download/clone this repo
- Open `Alia_blinky_esp32.ino`
- Select your board (step 1) and port
- Click Upload (→)

Hardware

LED Layout (41 WS2812B LEDs)

- **Prop 1-4:** 9 LEDs each (0-8, 9-17, 18-26, 27-35)
- **Tail:** 5 LEDs (36-40)

Pin Assignments

The code automatically detects your board and uses the correct GPIO pins. All boards use the same **physical pin positions** (TX, RX, SCK, MISO):

Board	Starboard (TX)	Port (RX)	Nose (SCK)	WS2812B (MISO)
Seeed XIAO RP2040	GPIO 0	GPIO 1	GPIO 2	GPIO 4
Adafruit QT Py RP2040	GPIO 20	GPIO 5	GPIO 6	GPIO 4
Seeed XIAO ESP32-S3	GPIO 43	GPIO 44	GPIO 8	GPIO 7
Adafruit QT Py ESP32-S3	GPIO 5	GPIO 16	GPIO 36	GPIO 37

Physical Connections:

- TX pin → Starboard navigation light (green)
- RX pin → Port navigation light (red)
- SCK pin → Nose navigation light (white)
- MISO pin → WS2812B data line

PCB Design Files

`alia_blinky_kicad.zip` contains complete KiCad project files for a custom PCB. The same PCB works with all supported boards - just swap the microcontroller!

Customization

Add Your Own Pattern

See [TUTORIAL.md](#) for a full walkthrough and several ready-to-paste examples. Quick summary:

1. Write pattern function in CUSTOM PATTERNS section (~line 1075)
2. Update NUM_PATTERNS constant (line 91)
3. Add case to switch statement (line 1268)
4. Add name to subModeNames array (line 1254)

Example:

```

void myPattern() {
  for (int i = 0; i < LED_COUNT; i++) {
    strip.setPixelColor(i, strip.Color(255, 0, 0)); // Red
  }
  strip.show();
  lights();
}

```

Adjust Settings

Brightness (line 96):

```

#define brightness 50 // 0-255 (lower for USB-A power)

```

Pattern Duration (line 147):

```

const int AUTO_CYCLE_DURATION = 10000; // milliseconds

```

Power Considerations

- **USB-C:** 2-3A (recommended)
- **USB-A:** 500mA (reduce brightness to 25)
- **External 5V:** 2A+ (best)

Pattern power draw:

- FLIGHT: 0.5-1.0A (sparse LEDs)
- RAINBOW: 2.0-2.5A (all LEDs)

Troubleshooting

LEDs brownout on USB-A power:

- Reduce brightness to 25
- Use USB-C or external 5V supply
- ESP32-S3 draws more power than RP2040 - use high-power USB port

Compilation errors:

- For RP2040: Ensure arduino-pico core v5.4.3+
- For ESP32: Ensure esp32 core v3.3.3+
- Verify Adafruit NeoPixel library installed
- Check you selected the correct board in Tools → Board

LEDs not working:

- Check serial monitor (115200 baud) - should show detected board type
- Verify correct board selected in Arduino IDE
- Check WS2812B connection (MISO pin - see Pin Assignments table)
- Verify 41 LEDs total
- ESP32-S3: Ensure "no psram" variant selected for QT Py

Wrong board detected:

- Serial monitor shows detected board on startup
- Ensure correct board selected in Tools → Board menu
- Code auto-detects based on compiler flags

Documentation

- [TUTORIAL.md](#) - Complete step-by-step guide
- [SOLDER.md](#) - Photo guide for soldering the microcontroller and gluing the diffuser covers
- [LICENSE](#) - MIT License
- **In-code comments** - Extensive documentation throughout

Credits

Created by: John Cohn, PhD **Original Date:** December 2024 **Updated:** January 2025 (Multi-board support added) **Contact:** jcohn@beta.team | johncohnvt@gmail.com

Developed with major assistance from Anthropic Claude.

Please reach out if you have comments, questions, or suggestions!

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